PROJECT SYNOPSIS

RetailVision: Local AI Analytics for Store-Camera Video

Detection, tracking, entrance counting, behaviour grouping, footfall forecasting and a private natural-language analyst

Field	Details
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1. Abstract

RetailVision is a locally run system that converts ordinary store-camera video into operating metrics: how many people are present, how many enter or leave through a doorway, how long they stay and how they move. A YOLOv8 detector and the ByteTrack tracker follow people through the footage; hand-written geometry counts line crossings; KMeans groups behaviour; the CLIP vision-language model reads whether a person carries a bag; an LSTM, chosen by backtest against simple baselines, forecasts footfall; and a locally hosted Qwen language model answers questions by writing guarded SQL. The system was evaluated on a public mall dataset and two supplied store clips. Occupancy error was about 14%, doorway crossings reached about 70% precision and 71% recall, and bag detection reached 74% accuracy on large crops. All processing and data stay on the user's machine.

2. Introduction and Problem Statement

Retailers decide staffing, opening hours and layout on estimates of footfall and customer behaviour, yet most stores only have a camera feed that nobody analyses. Manual counting is slow and inconsistent, and commercial analytics products are typically proprietary, subscription-based and cloud-hosted, which raises both cost and privacy concerns when video of customers leaves the premises.

Turning raw video into dependable numbers is hard in practice. Shoppers occlude each other, groups walk as one blob, cameras are often low frame rate and low resolution, and there is rarely labelled data for the specific store. A system that is useful must therefore be measured honestly, state where it fails, and avoid presenting guesses as facts. This project builds such a system end to end and documents what it can and cannot claim.

3. Objectives

- **Occupancy:** detect and track people in video and report how many are visible over time.
- **Entrance counting:** count people crossing a user-drawn line, in and out, with a stated accuracy.
- **Behaviour:** report dwell time, location and behaviour groups for each tracked person.
- **Forecasting:** predict footfall and select the method by honest backtesting, not by assumption.
- **Private analyst:** let non-technical users query results in plain English without sending data off the machine.
- **Measured claims:** evaluate each component against ground truth and document limitations.

4. Technologies and Algorithms

Ten components are used, each with a defined role. Three families appear: pretrained neural networks used unchanged, models fitted to the user's own data, and hand-written rules.

Component	Method	Role in RetailVision	Learns from
YOLOv8s	Convolutional object detector	Finds every person in each frame (1280 px, confidence 0.15)	COCO (pretrained)
ByteTrack	Kalman filter + Hungarian matching	Links boxes over time into one track per person; keeps weak boxes	Nothing
Line crossing	Signed-distance geometry	Counts in/out events with margin, segment test and reversal cancelling	Nothing
Fragment merge	Greedy heuristic	Stitches broken tracks, with a coverage guard	Nothing
KMeans (k = 3)	Unsupervised clustering	Groups people as passer-by, browser or engaged	The video itself
CLIP ViT-B/32	Vision-language, zero-shot	Decides if a large person crop is carrying a bag	Pretrained
LSTM	Recurrent neural network	Forecasts hourly footfall; used only if it wins a backtest	Store history
Seasonal / last value	Baselines	Same hour yesterday; the last observed value	Nothing
Qwen2.5-1.5B	Transformer language model	Writes one SQL SELECT per question and explains the rows	Pretrained
SQL guard	Database authorizer and limits	Allow-list of 4 tables, read-only, 5 s limit, 50 rows	Nothing

4.1 Why these choices

Each choice was made on a measured reason, and each has a stated cost. The table also records the alternative that was not taken.

Decision	Chosen	Instead of	Reason	Cost
Detector	YOLOv8s at 1280 px	Nano model at 640 px	Error 4.1 vs 13.2 people per frame	Slower
Tracker	ByteTrack	DeepSORT, BoT-SORT	No second model; keeps weak boxes	Identity switches in crowds
Groups	KMeans	Fixed time thresholds	Needs no labels	Names are relative per video
Attributes	CLIP zero-shot	Train a classifier	No labelled data; 74% on large crops	Weak on small people
Forecast	LSTM chosen by backtest	Baselines only	Can learn a daily shape	Needs days of data
Analyst	Local Qwen 1.5B	Cloud language model	Privacy, no external API	About 90% on new wording
Storage	SQLite	Server database	One file, no set-up	Single machine

4.2 How the families differ

Detection answers who is in one frame and has no memory; tracking adds time so a person keeps one identity, so detection errors are per-frame while tracking errors are identity errors. YOLO, CLIP and Qwen are trained by others and used unchanged; only KMeans (per video) and the LSTM (per store) are fitted to the user's data; tracking, counting and merging are rules. KMeans needs no labels, CLIP replaces labels with sentences, and the LSTM is supervised by its own series. The LSTM predicts a number, whereas the language model generates text, which is why its output is executed through a guard rather than trusted.

5. System Design and Methodology

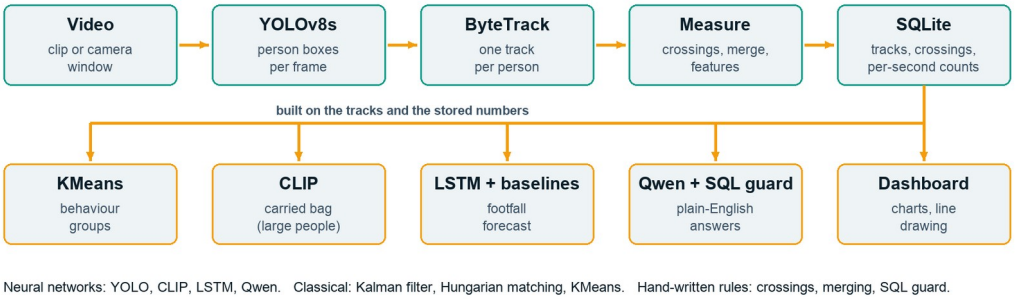


Figure 1. System pipeline: detection and tracking feed measurements into one database; every other component reads from it.

5.1 Detection and tracking

Each processed frame goes through YOLOv8s, keeping only the person class. The boxes are passed to ByteTrack, which predicts each track with a Kalman filter, matches strong detections by overlap using the Hungarian algorithm, and then matches leftover tracks to weak detections, which are usually partly hidden people. Detection and association are run separately because the library's all-in-one call returns only confirmed tracks and hid roughly 40% of people at low frame rate. Occupancy is the mean number of detections per second.

5.2 Counting, merging and behaviour

A counting line is two points in frame fractions. For each track point the signed distance to the line is computed with a cross product; a crossing is recorded only when a track moves from beyond a margin on one side to beyond it on the other, within the segment, and a crossing undone within three seconds is cancelled as jitter or an identity switch. Per-person dwell time, path length, speed and stationary ratio are then standardised and clustered by KMeans; clusters are named by their centres (fastest = passer-by, longest dwell of the rest = engaged-shopper).

5.3 Attributes, forecast and analyst

CLIP scores the largest crop of a person (at least 180 px tall) against “carrying a bag or basket” minus “carrying nothing”. Hourly occupancy feeds a one-layer LSTM (32 units, six-hour window); a backtest on the last 20% compares it with last-value and same-hour-yesterday baselines and the lowest-error method is used. For questions, Qwen2.5-1.5B writes SQL from a schema, rules and 17 examples; deterministic fixes remove known mistakes, an error-aware retry repairs missing columns, and the SQL guard enforces read-only access before the rows are explained in a sentence.

6. Implementation

The backend is Python with FastAPI; analysis uses PyTorch, Ultralytics, OpenCV, scikit-learn and Hugging Face Transformers, and results are stored in SQLite (videos, tracks, crossings, per-second counts, jobs, counting lines). Uploads return at once with a job identifier; a single worker processes a persistent queue that survives restarts, and a rolling monitor can analyse a configured camera in back-to-back windows. The frontend is HTML and JavaScript with Chart.js: key figures, an upload panel, a canvas on which the user draws the counting line, forecast and behaviour charts, a zone heat map and a chat panel that shows the SQL it executed and marks failures as not executed. Optional shared-key authentication, rate limits, data retention and a delete-all control are included. Quality is protected by 84 automated tests, including real-browser tests and a deterministic end-to-end test with a scripted detector, run in continuous integration.

7. Results and Evaluation

7.1 Datasets and protocol

Three sources were used. The public CUHK Mall dataset (2,000 CCTV frames with per-frame ground-truth people counts, research use only) measured detection and occupancy. Two clips supplied by the user, a

pharmacy interior (111 s) and a store entrance with automatic doors (121 s), provided real store footage. For the entrance clip, a candidate pool of 102 crossing events was assembled from the production settings plus a looser detector pass to expose missed crossings; each candidate was labelled by one annotator as a true first crossing, a duplicate of the same person, a wrong direction, or unsure, and precision, recall and count error were computed from those labels. Bag labels were assigned to 53 person crops. The language model was tested on a tuned question set and on 12 questions written afterwards.

7.2 Findings

Measure	Setup	Result	Strength of evidence
Detection error	200 frames, public mall dataset with ground-truth counts	4.1 people/frame (13%) vs 13.2 (42%) with defaults	Good
Occupancy error	200 s mall clip, per second	3.8 people (about 14%)	Good
Doorway crossings	2-minute entrance clip, 102 candidates hand-labelled	Precision 70%, recall 71%; 57 reported vs about 52 real	Indicative (one clip, one annotator)
Carried bag	53 crops, people 188–295 px tall	74% accuracy vs 51% baseline	Indicative (about ±12 points)
Analyst	36 tuned and 12 unseen questions	100% tuned (lenient); 92% on unseen	Indicative
Soak test	30 minutes, looping clip, isolated server	135 windows, 0 errors, memory +4.4%	Good (memory noisy)

Three engineering findings mattered more than any single model choice: running detection and association separately cut occupancy error from 12.2 to 3.8 people; cancelling same-track reversals raised crossing precision from 58% to 70% at some cost in recall; and CLIP's usefulness depended on crop size, falling below a trivial baseline on small crops. Crossing reversal cancelling removed 30 events, of which 17 were correct removals and 9 were real crossings, so it trades recall for precision.

8. Limitations and Future Scope

- **Identity:** “people tracked” is inflated by identity switches in crowds and is labelled an estimate; occupancy and crossings are the dependable figures.
- **Staff versus customer:** never validated, because no uniform is visible in the test footage; the feature is off.
- **Forecast:** needs days of history; on short data it only matched a simple baseline, and no real-store benefit has been shown.
- **Scale of evidence:** crossing and bag results come from one clip and one annotator; larger labelled sets are needed.
- **Streaming:** the monitor is near-continuous with gaps while each window is analysed, not frame-by-frame real time.

Future work: a hand-tracked identity ground truth to tune merging; an appearance re-identification model if it can be validated; named store areas instead of a grid; accuracy gates in continuous integration; multi-day data collection to evaluate the forecast; and per-user accounts with HTTPS for deployment.

9. Conclusion

RetailVision shows that a useful store-analytics pipeline can run entirely on a single machine using open models, provided each component is measured and its limits are stated. Occupancy and doorway counting are practical today; behaviour grouping and the analyst are helpful but approximate; the forecast and staff recognition are not yet proven. The main contribution is the combination of working components with an evaluation habit that reports what is, and is not, known.

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